

# ETF Forecaster - Methodology and Results

Generated 2026-09-04 02:21 - mo-etf-forecaster - universe of 40 tickers, horizons 1-5 days

## 1. Methodology (design of record)

**Objective.** For each of 40 ETFs/indices, forecast the probability that the close in  $h$  trading days ( $h = 1..5$ ) is above today's close, plus a conformal quantile distribution of the  $h$ -day log return, and maintain an audited out-of-sample track record.

**Data.** Full-history daily bars (yfinance) for the universe plus the CBOE volatility complex (VIX, VIX9D, VIX3M, VIX6M, VVIX, SKEW, VXN, VXD, OVX, GVZ) and 10 cross-asset series; 14 fast FRED series (yields, curve, credit OAS, dollar, oil, NFCI, Fed balance sheet, SOFR); 24 slow macro series pulled as point-in-time ALFRED vintages (CPI, PCE, INDPRO, capacity utilization, payrolls, claims, M2, GDP, housing, sentiment, NBER dates); Shiller S&P 500 earnings/CAPE; per-ticker 30-day implied and historical volatility from IBKR; and a daily option-chain snapshot out to 120 DTE from which 25-delta risk reversal, butterfly, ATM term structure and a dealer-gamma proxy are computed (this block accrues history forward from first snapshot).

**Point-in-time discipline.** Slow macro features are computed from the series exactly as known on each historical date via ALFRED vintages (fallback: conservative publication lags). Weekly FRED series are lagged one week. Korean closes are same-day usable for US targets but US closes are lagged one day for Korean targets. Weekly/monthly chart patterns only become visible the first daily bar after their (completed) higher-timeframe bar ends. Shiller earnings are lagged three months.

**Features.** ~690 columns per ticker in 12 blocks: PRICE (always on; returns, gaps, range-vol estimators, last-60-candle block), TECH (20+ indicators), PATTERN\_D1 and PATTERN\_HTF (16 chart-pattern detectors - double/triple tops and bottoms, head and shoulders, channels, S/R, wedges, flags, trendlines and breaks - run on daily, weekly and monthly bars, converted to per-bar features with direction decay, trigger flags, ATR distances and cross-timeframe agreement), REGIME (ADX state, HH/HL market structure, variance ratio, MA posture, vol terciles, leak-free Markov-switching probability), VOLOPT (VIX complex, term slopes, IBKR IV, VRP, HAR-RV, EWMA/GARCH), SKEW (chain-derived), MACRO\_FAST, MACRO\_SLOW (vintage-aligned), EARNINGS, CROSS (session-aligned lead-lag, relative strength, beta, group momentum rank), SEASON (calendar and event days).

**Search.** A staged include/exclude ablation over the 11 toggleable blocks per (ticker-group, horizon): stage 1 screens each block against a PRICE-only floor; stage 2 greedy forward/backward selection; stage 3 exhaustive  $2^k$  over the shortlist; stage 4 model/hyperparameter search (logistic, elastic-net, LightGBM, XGBoost, CatBoost, RF/ET/HGB, LSTM/TCN over the candle tensor; Optuna TPE in full mode) including the lag-window  $X$  in  $\{10, 20, 40, 60\}$ ; stage 5 stacking of the top configurations with isotonic calibration. All selection is nested in the development slice (first 60% of history). Every evaluated cell is persisted with its running trial count.

**Validation.** Purged, embargoed, expanding walk-forward: training data for a refit forecasting date  $t$  ends  $h$  bars before  $t$  (both purge and embargo for overlapping  $h$ -day labels). The out-of-sample track record consists exclusively of predictions dated after the development slice; champion switching along the way uses only trailing OOS data available at that moment (sticky rule). Magnitude quantiles are conformalized (CQR) on a held-out calibration slice targeting 80% coverage for the  $q_{10}$ - $q_{90}$  band. Reported results carry Benjamini-Hochberg FDR flags and binomial  $p$ -values; hit rates should be read net of the recorded trial counts.

**Honest expectations.** Daily equity-index direction is a low-signal problem: a realistic sustained OOS hit rate is 52-56%, not 70%. VIX-family tickers score much higher hit rates for structural reasons (persistent downward drift of VIXY, mean reversion of VIX), which the probability calibration reflects. The chain-derived SKEW block has essentially no history yet and will only earn a place in the ablation as snapshots accrue.

## 2. Data dictionary and coverage

Daily bars (yfinance), full available history:

| ticker    | rows  | start      | end        |
|-----------|-------|------------|------------|
| SPY       | 8457  | 1993-01-29 | 2026-09-03 |
| QQQ       | 6915  | 1999-03-10 | 2026-09-03 |
| IWM       | 6607  | 2000-05-26 | 2026-09-03 |
| MDY       | 7886  | 1995-05-04 | 2026-09-03 |
| DIA       | 7201  | 1998-01-20 | 2026-09-03 |
| XLK       | 6967  | 1998-12-22 | 2026-09-03 |
| XLF       | 6967  | 1998-12-22 | 2026-09-03 |
| XLE       | 6967  | 1998-12-22 | 2026-09-03 |
| XLB       | 6967  | 1998-12-22 | 2026-09-03 |
| XLI       | 6967  | 1998-12-22 | 2026-09-03 |
| XLV       | 6967  | 1998-12-22 | 2026-09-03 |
| XLY       | 6967  | 1998-12-22 | 2026-09-03 |
| XLP       | 6967  | 1998-12-22 | 2026-09-03 |
| XLC       | 2064  | 2018-06-19 | 2026-09-03 |
| XLU       | 6967  | 1998-12-22 | 2026-09-03 |
| XLRE      | 2742  | 2015-10-08 | 2026-09-03 |
| ^VIX      | 9237  | 1990-01-02 | 2026-09-03 |
| ^VVIX     | 4940  | 2007-01-03 | 2026-09-03 |
| VIXY      | 3940  | 2011-01-04 | 2026-09-03 |
| SOXX      | 6323  | 2001-07-13 | 2026-09-03 |
| SMH       | 6602  | 2000-06-05 | 2026-09-03 |
| MU        | 10647 | 1984-06-01 | 2026-09-03 |
| 000660.KS | 6667  | 2000-01-04 | 2026-09-03 |
| 005930.KS | 6667  | 2000-01-04 | 2026-09-03 |
| EWG       | 7666  | 1996-03-18 | 2026-09-03 |
| EWQ       | 7666  | 1996-03-18 | 2026-09-03 |
| EWJ       | 7666  | 1996-03-18 | 2026-09-03 |
| EWU       | 7666  | 1996-03-18 | 2026-09-03 |
| EWC       | 7666  | 1996-03-18 | 2026-09-03 |
| EEM       | 5886  | 2003-04-14 | 2026-09-03 |
| FXI       | 5511  | 2004-10-08 | 2026-09-03 |
| EWY       | 6617  | 2000-05-12 | 2026-09-03 |
| EWT       | 6588  | 2000-06-23 | 2026-09-03 |
| INDA      | 3667  | 2012-02-03 | 2026-09-03 |

| ticker   | rows  | start      | end        |
|----------|-------|------------|------------|
| EWZ      | 6574  | 2000-07-14 | 2026-09-03 |
| EWV      | 7666  | 1996-03-18 | 2026-09-03 |
| ^GSPTSE  | 11847 | 1979-06-29 | 2026-09-03 |
| XFN.TO   | 6387  | 2001-03-29 | 2026-09-03 |
| XEG.TO   | 6391  | 2001-03-23 | 2026-09-03 |
| XMA.TO   | 5193  | 2005-12-28 | 2026-09-03 |
| ^VIX     | 9237  | 1990-01-02 | 2026-09-03 |
| ^VIX9D   | 3909  | 2011-01-03 | 2026-09-03 |
| ^VIX3M   | 5034  | 2006-07-17 | 2026-09-03 |
| ^VIX6M   | 4666  | 2008-01-02 | 2026-09-03 |
| ^VVIX    | 4940  | 2007-01-03 | 2026-09-03 |
| ^SKEW    | 9162  | 1990-01-02 | 2026-09-03 |
| ^VXN     | 6442  | 2001-01-23 | 2026-09-03 |
| ^RVX     | 0     | nan        | nan        |
| ^VXD     | 5190  | 2005-11-22 | 2026-09-03 |
| ^OVX     | 4861  | 2007-05-10 | 2026-09-03 |
| ^GVZ     | 4593  | 2008-06-03 | 2026-09-03 |
| ^TNX     | 16156 | 1962-01-02 | 2026-09-03 |
| ^TYX     | 12414 | 1977-02-15 | 2026-09-03 |
| DX-Y.NYB | 14136 | 1971-01-04 | 2026-09-03 |
| CL=F     | 6536  | 2000-08-23 | 2026-09-03 |
| GC=F     | 6527  | 2000-08-30 | 2026-09-03 |
| HG=F     | 6532  | 2000-08-30 | 2026-09-03 |
| TLT      | 6064  | 2002-07-30 | 2026-09-03 |
| HYG      | 4882  | 2007-04-11 | 2026-09-03 |
| LQD      | 6064  | 2002-07-30 | 2026-09-03 |
| RSP      | 5874  | 2003-05-01 | 2026-09-03 |

Other artifacts (vol indices, macro, vintages, IBKR IV, skew):

| artifact        | rows  | start      | end        |
|-----------------|-------|------------|------------|
| calendar/events | 8457  | 1993-01-29 | 2026-09-03 |
| cross/CL_F      | 6536  | 2000-08-23 | 2026-09-03 |
| cross/DX-Y_NYB  | 14136 | 1971-01-04 | 2026-09-03 |
| cross/GC_F      | 6527  | 2000-08-30 | 2026-09-03 |
| cross/HG_F      | 6532  | 2000-08-30 | 2026-09-03 |
| cross/HYG       | 4882  | 2007-04-11 | 2026-09-03 |
| cross/LQD       | 6064  | 2002-07-30 | 2026-09-03 |

| artifact     | rows  | start      | end        |
|--------------|-------|------------|------------|
| cross/RSP    | 5874  | 2003-05-01 | 2026-09-03 |
| cross/TLT    | 6064  | 2002-07-30 | 2026-09-03 |
| cross/TNX    | 16156 | 1962-01-02 | 2026-09-03 |
| cross/TYX    | 12414 | 1977-02-15 | 2026-09-03 |
| fred/fast    | 9568  | 1990-01-02 | 2026-09-03 |
| ibkr/DIA_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/DIA_iv  | 3768  | 2011-09-07 | 2026-09-02 |
| ibkr/EEM_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EEM_iv  | 3768  | 2011-09-07 | 2026-09-02 |
| ibkr/EWC_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EWC_iv  | 3768  | 2011-09-07 | 2026-09-02 |
| ibkr/EWG_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EWG_iv  | 3767  | 2011-09-07 | 2026-09-02 |
| ibkr/EWJ_hv  | 3758  | 2011-09-06 | 2026-09-01 |
| ibkr/EWJ_iv  | 3768  | 2011-09-06 | 2026-09-02 |
| ibkr/EWQ_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EWQ_iv  | 3756  | 2011-09-07 | 2026-09-02 |
| ibkr/EWT_hv  | 3758  | 2011-09-06 | 2026-09-01 |
| ibkr/EWT_iv  | 3768  | 2011-09-06 | 2026-09-02 |
| ibkr/EWU_hv  | 3757  | 2011-09-06 | 2026-09-01 |
| ibkr/EWU_iv  | 3689  | 2011-09-07 | 2026-09-02 |
| ibkr/WWW_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/WWW_iv  | 3768  | 2011-09-07 | 2026-09-02 |
| ibkr/EWY_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EWY_iv  | 3767  | 2011-09-07 | 2026-09-02 |
| ibkr/EWZ_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/EWZ_iv  | 3767  | 2011-09-07 | 2026-09-02 |
| ibkr/FXI_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/FXI_iv  | 3767  | 2011-09-07 | 2026-09-02 |
| ibkr/INDA_hv | 3636  | 2012-03-19 | 2026-09-01 |
| ibkr/INDA_iv | 3247  | 2013-07-18 | 2026-09-02 |
| ibkr/IWM_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/IWM_iv  | 3768  | 2011-09-07 | 2026-09-02 |
| ibkr/MDY_hv  | 3757  | 2011-09-07 | 2026-09-01 |
| ibkr/MDY_iv  | 3757  | 2011-09-07 | 2026-09-02 |
| ibkr/MU_hv   | 3757  | 2011-09-07 | 2026-09-01 |

| artifact        | rows | start      | end        |
|-----------------|------|------------|------------|
| ibkr/MU_iv      | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/QQQ_hv     | 3758 | 2011-09-06 | 2026-09-01 |
| ibkr/QQQ_iv     | 3769 | 2011-09-06 | 2026-09-02 |
| ibkr/SMH_hv     | 3652 | 2012-02-06 | 2026-09-01 |
| ibkr/SMH_iv     | 3691 | 2011-12-22 | 2026-09-02 |
| ibkr/SOXX_hv    | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/SOXX_iv    | 3690 | 2011-09-07 | 2026-09-02 |
| ibkr/SPY_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/SPY_iv     | 3767 | 2011-09-07 | 2026-09-02 |
| ibkr/VIXY_hv    | 3750 | 2011-09-06 | 2026-09-01 |
| ibkr/VIXY_iv    | 3763 | 2011-09-06 | 2026-09-02 |
| ibkr/XLB_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLB_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLC_hv     | 2025 | 2018-08-01 | 2026-09-01 |
| ibkr/XLC_iv     | 2060 | 2018-06-22 | 2026-09-02 |
| ibkr/XLE_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLE_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLF_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLF_iv     | 3766 | 2011-09-07 | 2026-09-02 |
| ibkr/XLI_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLI_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLK_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLK_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLP_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLP_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLRE_hv    | 2703 | 2015-11-19 | 2026-09-01 |
| ibkr/XLRE_iv    | 2719 | 2015-10-27 | 2026-09-02 |
| ibkr/XLU_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLU_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLV_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLV_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| ibkr/XLY_hv     | 3757 | 2011-09-07 | 2026-09-01 |
| ibkr/XLY_iv     | 3768 | 2011-09-07 | 2026-09-02 |
| shiller/shiller | 1845 | 1871-01-31 | 2024-09-30 |
| skew/DIA        | 2    | 2026-09-02 | 2026-09-03 |
| skew/EEM        | 2    | 2026-09-02 | 2026-09-03 |

| artifact | rows | start      | end        |
|----------|------|------------|------------|
| skew/EWC | 2    | 2026-09-02 | 2026-09-03 |

Fast FRED series: DGS3MO, DGS2, DGS10, T10Y2Y, T10Y3M, DFII10, T10Y1E, BAMLH0A0HYM2, BAMLC0A0CM, DTWEXBGS, DCOILWTICO, NFCI, WALCL, SOFR.

ALFRED vintage series: CPIAUCSL, CPILFESL, PCEPI, PCEPILFE, MICH, INDPRO, IPMAN, TCU, NEWORDER, HOUST, PERMIT, UNRATE, U6RATE, PAYEMS, ICSA, CCSA, SAHMREALTIME, M2SL, WM2NS, M2REAL, GDPC1, UMCSENT, USREC, CP.

### 3. Feature blocks

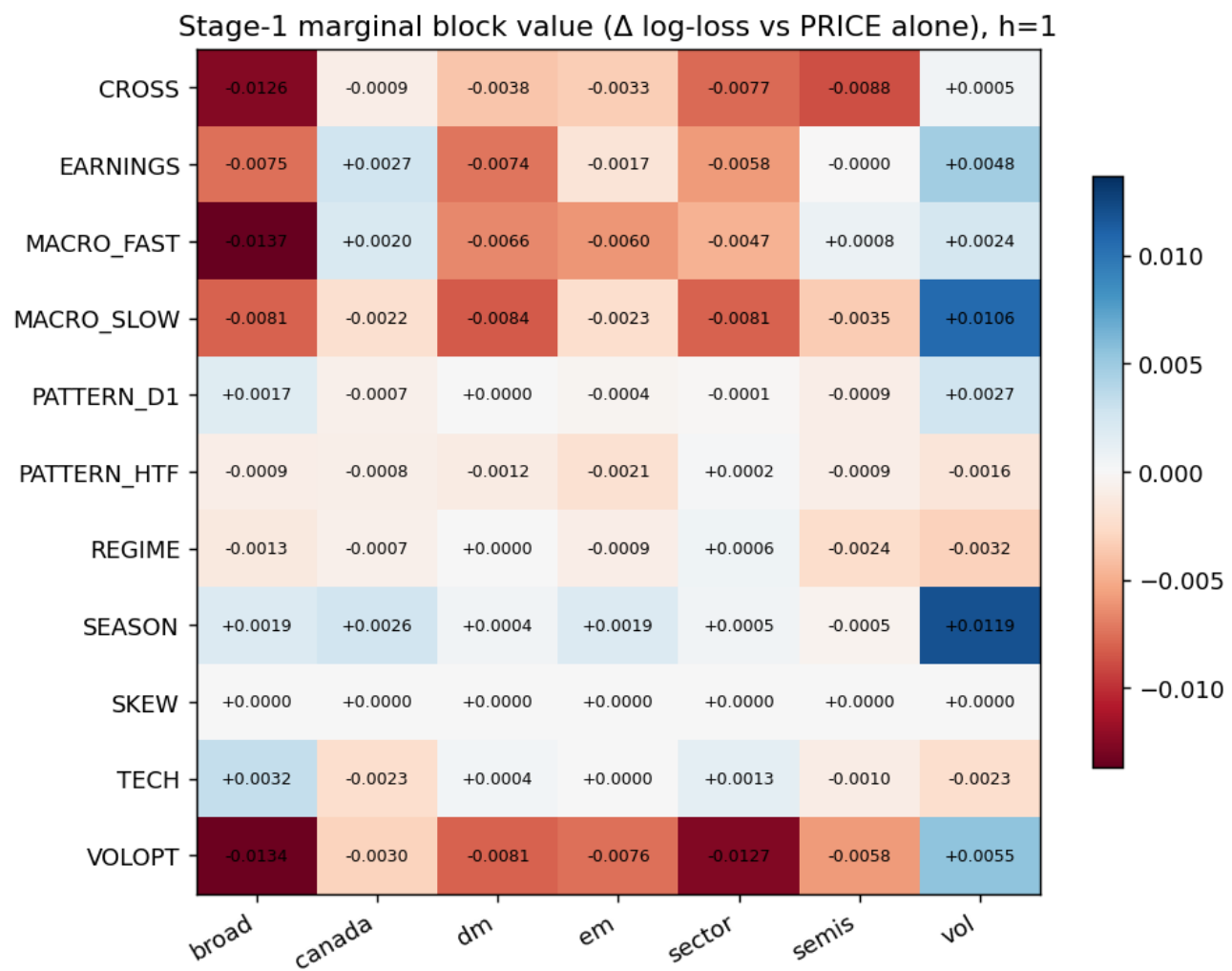
| block       | prefix | toggleable | columns (SPY) | description                                          |
|-------------|--------|------------|---------------|------------------------------------------------------|
| PRICE       | px_    | False      | 112           | returns, gaps, range-vol estimators, last-X candles  |
| TECH        | tech_  | True       | 39            | top-20 technical indicators                          |
| PATTERN_D1  | patd_  | True       | 53            | chart patterns on daily bars                         |
| PATTERN_HTF | path_  | True       | 105           | chart patterns on weekly/monthly bars + agreement    |
| REGIME      | reg_   | True       | 15            | trend/range state, market structure, Markov regime   |
| VOLOPT      | vol_   | True       | 35            | VIX complex, term slopes, IBKR IV, HAR/GARCH         |
| SKEW        | skw_   | True       | 15            | chain-derived rr25/bf25/term/GEX (short history)     |
| MACRO_FAST  | mf_    | True       | 58            | daily FRED: yields, curve, credit, dollar, oil       |
| MACRO_SLOW  | ms_    | True       | 132           | vintage-aligned CPI, INDPRO, UNRATE, claims, M2, ... |
| EARNINGS    | earn_  | True       | 9             | Shiller earnings growth, CAPE, Fed-model spread      |
| CROSS       | x_     | True       | 102           | session-aligned cross-asset lead-lag and ranks       |
| SEASON      | cal_   | True       | 19            | day-of-week, turn-of-month, OPEX, FOMC, releases     |

### 4. Grid-search design and budgets

Horizons: [1, 2, 3, 4, 5]. Walk-forward: min\_train 1000 bars, refit every 21 (full) / 63 (fast) bars, embargo = horizon, calibration slice 15%. Ablation candidates: TECH, PATTERN\_D1, PATTERN\_HTF, REGIME, VOLOPT, SKEW, MACRO\_FAST, MACRO\_SLOW, EARNINGS, CROSS, SEASON. Greedy min gain 0.0005, exhaustive cap  $2^6$ . Optuna trials per cell (full): 40. Window grid: [10, 20, 40, 60]. Champion metric log\_loss with sticky margin 0.002; BH-FDR at 10%.

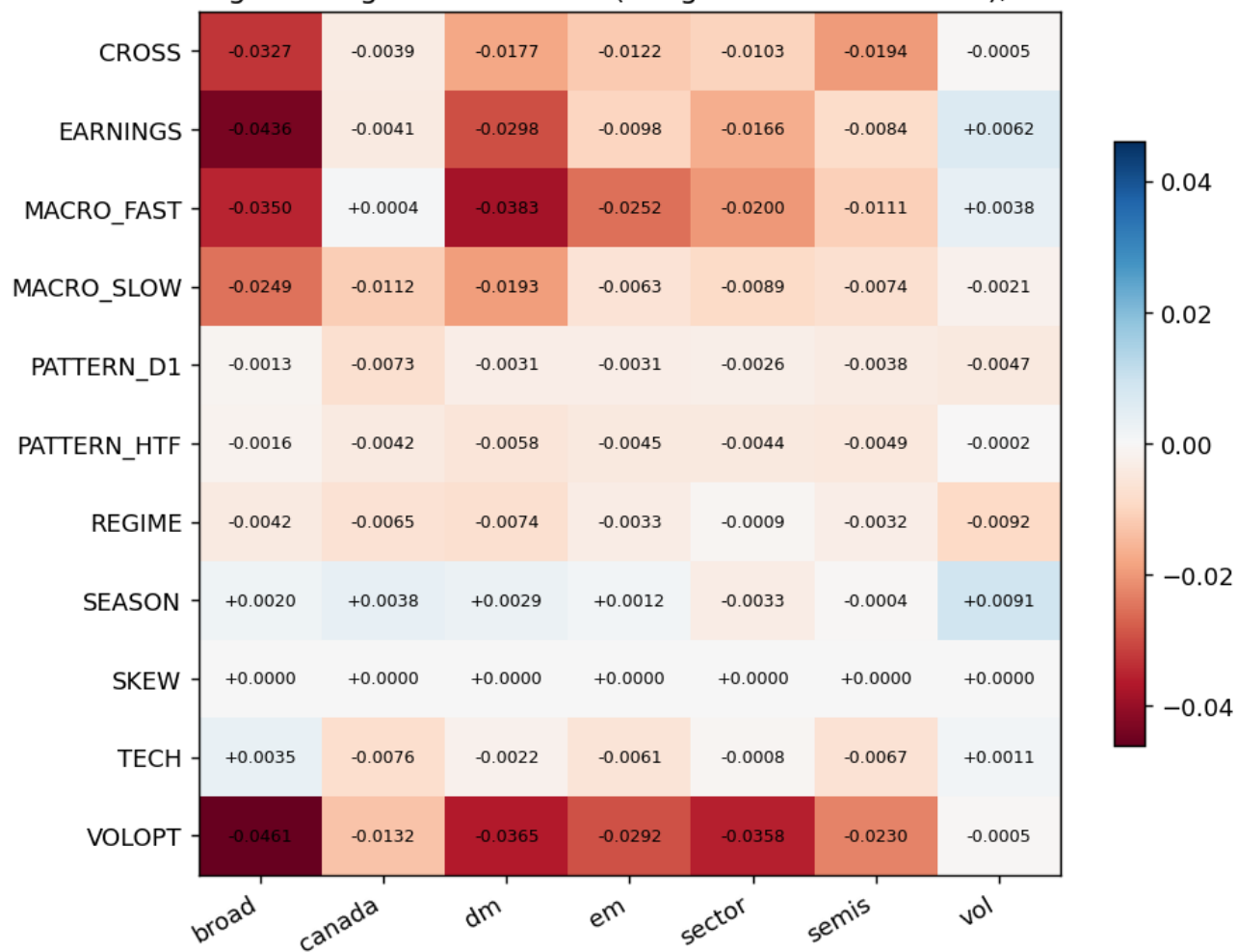
Ablation cells evaluated to date: **1020** across 7 groups x 3 horizons ({'s2\_greedy': 346, 's4\_model': 267, 's1\_screen': 252, 's3\_exhaustive': 155}).

## 5. Ablation results - which blocks earn their keep

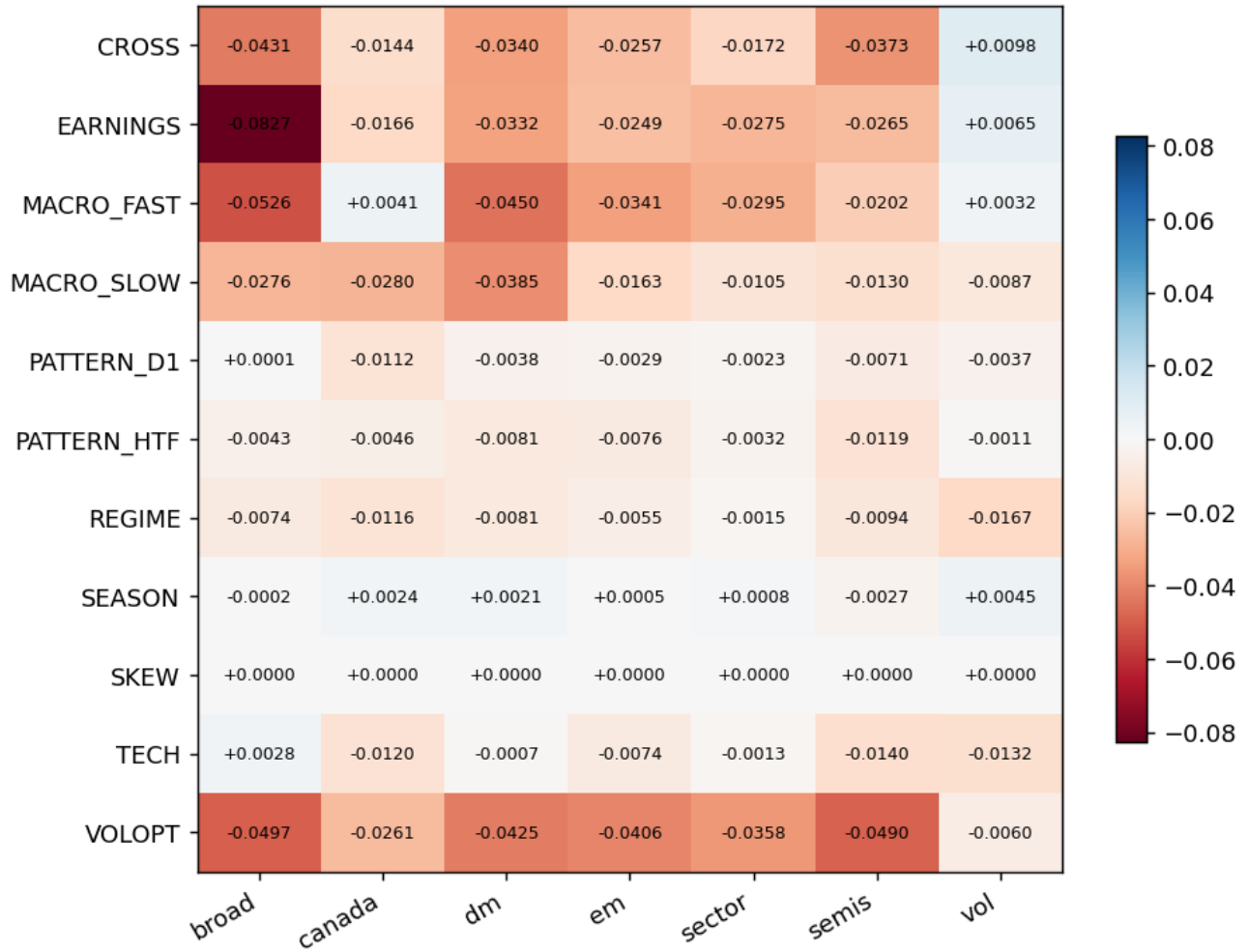




Stage-1 marginal block value ( $\Delta$  log-loss vs PRICE alone),  $h=3$



Stage-1 marginal block value ( $\Delta$  log-loss vs PRICE alone),  $h=5$



Winning block sets (stage-3 exhaustive):

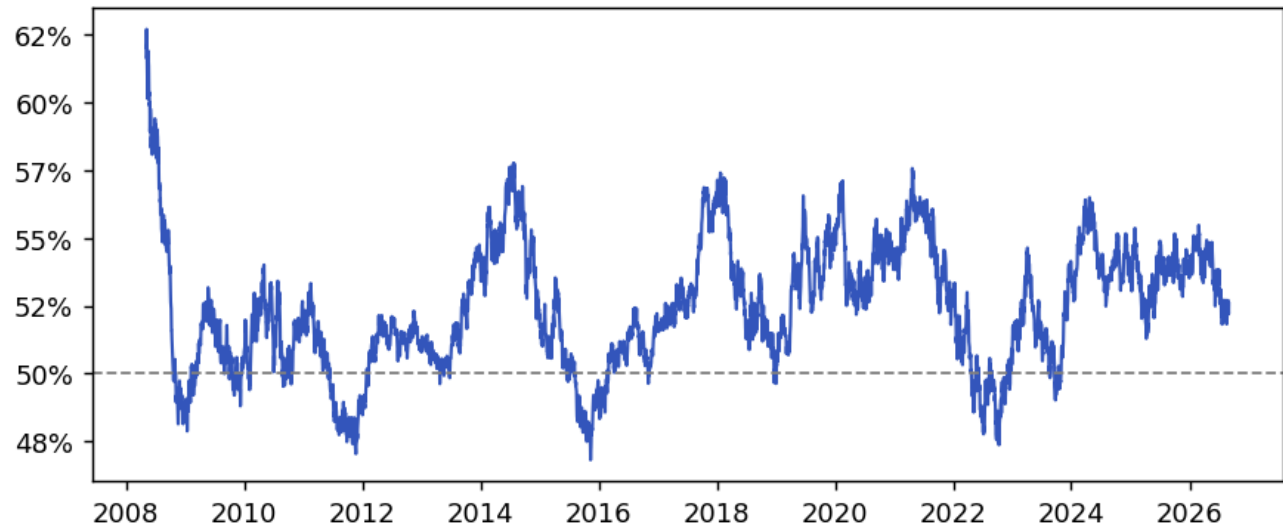
| group  | horizon | blocks                        | log_loss | hit_rate | n     | n_trials_so_far |
|--------|---------|-------------------------------|----------|----------|-------|-----------------|
| broad  | 1       | SEASON,TECH                   | 0.7086   | 0.5226   | 18457 | 22              |
| broad  | 3       | TECH                          | 0.713    | 0.5356   | 18452 | 2               |
| broad  | 5       | PATTERN_D1,TECH               | 0.7215   | 0.5381   | 18447 | 13              |
| canada | 1       | EARNINGS,MACRO_FAST,SEASON    | 0.6973   | 0.5459   | 9500  | 29              |
| canada | 3       | SEASON                        | 0.707    | 0.5344   | 9496  | 12              |
| canada | 5       | MACRO_FAST,PATTERN_HTF,SEASON | 0.7204   | 0.5446   | 9492  | 25              |
| dm     | 1       | PATTERN_D1,SEASON,TECH        | 0.6986   | 0.522    | 18485 | 36              |
| dm     | 3       | SEASON                        | 0.7035   | 0.5351   | 18480 | 12              |
| dm     | 5       | SEASON                        | 0.7141   | 0.53     | 18470 | 12              |
| em     | 1       | PATTERN_HTF,SEASON            | 0.703    | 0.5224   | 19226 | 15              |
| em     | 3       | SEASON                        | 0.7126   | 0.5218   | 19219 | 12              |
| em     | 5       | SEASON                        | 0.7253   | 0.5171   | 19205 | 12              |
| sector | 1       | REGIME,TECH                   | 0.6974   | 0.5197   | 29493 | 15              |

| group  | horizon | blocks                                  | log_loss | hit_rate | n     | n_trials_so_far |
|--------|---------|-----------------------------------------|----------|----------|-------|-----------------|
| sector | 3       | PRICE_only                              | 0.7018   | 0.5276   | 29484 | 1               |
| sector | 5       | SEASON                                  | 0.7082   | 0.5307   | 29475 | 12              |
| semis  | 1       | MACRO_FAST                              | 0.706    | 0.5265   | 13721 | 8               |
| semis  | 3       | PRICE_only                              | 0.7125   | 0.5135   | 13716 | 1               |
| semis  | 5       | PRICE_only                              | 0.718    | 0.511    | 13711 | 1               |
| vol    | 1       | CROSS,EARNINGS,MACRO_SLOW,SEASON,VOLOPT | 0.6943   | 0.5657   | 5833  | 56              |
| vol    | 3       | EARNINGS,MACRO_FAST,SEASON              | 0.705    | 0.5744   | 5830  | 29              |
| vol    | 5       | CROSS,EARNINGS,MACRO_FAST,SEASON        | 0.7174   | 0.5782   | 5827  | 39              |

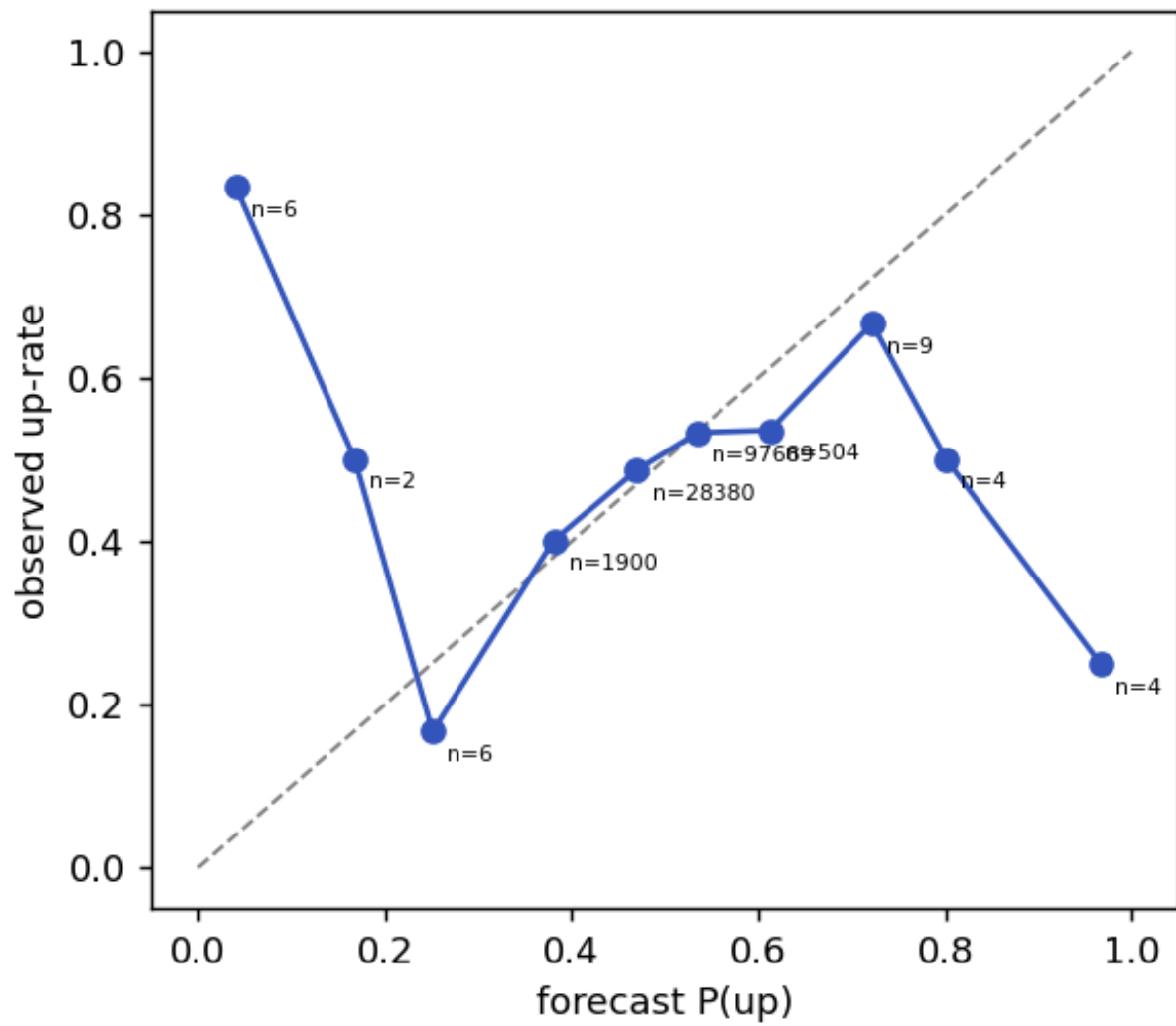
## 6. Out-of-sample performance

Track record: **642,354** matured champion-series OOS predictions across 40 tickers, from 2008-03-06 to 2026-09-01. Every prediction ever made is stored in predictions\_history.parquet with its realized outcome.

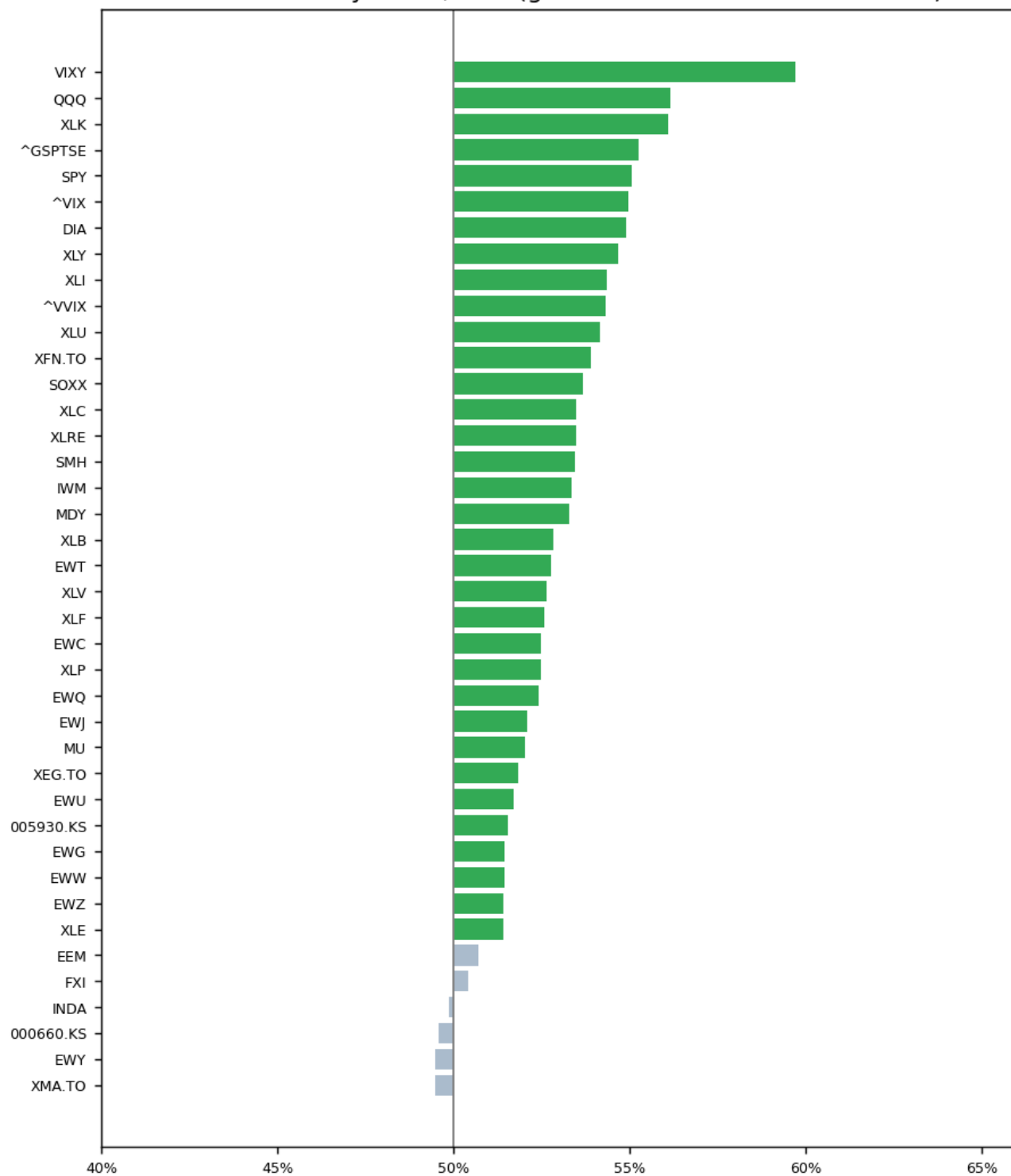
126-day rolling OOS hit rate pooled across tickers,  $h=1$



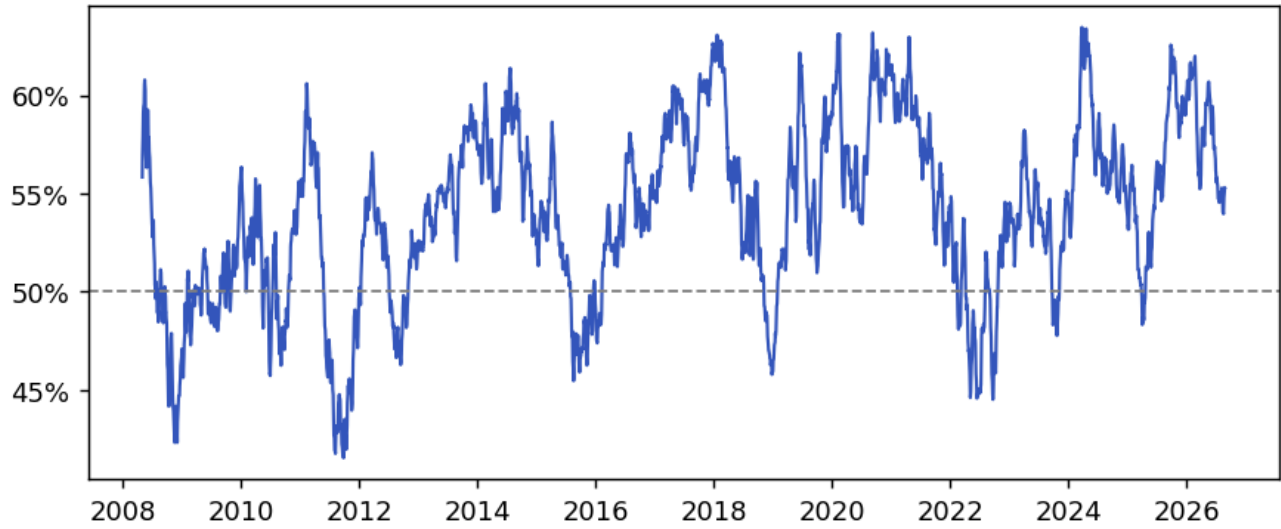
Reliability, champion series, h=1 (128504 OOS preds)



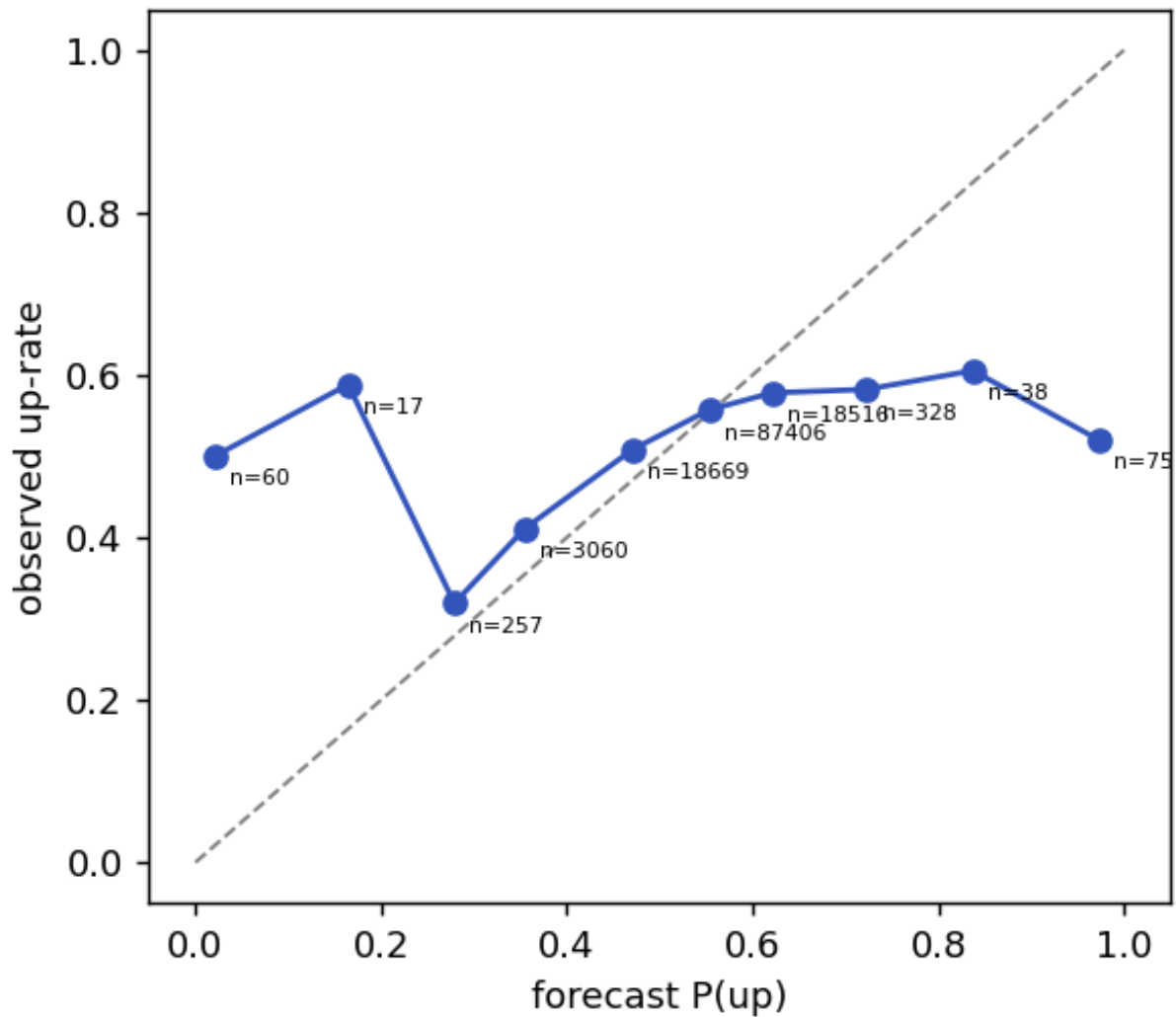
OOS hit rate by ticker, h=1 (green = survives BH-FDR at 10%)



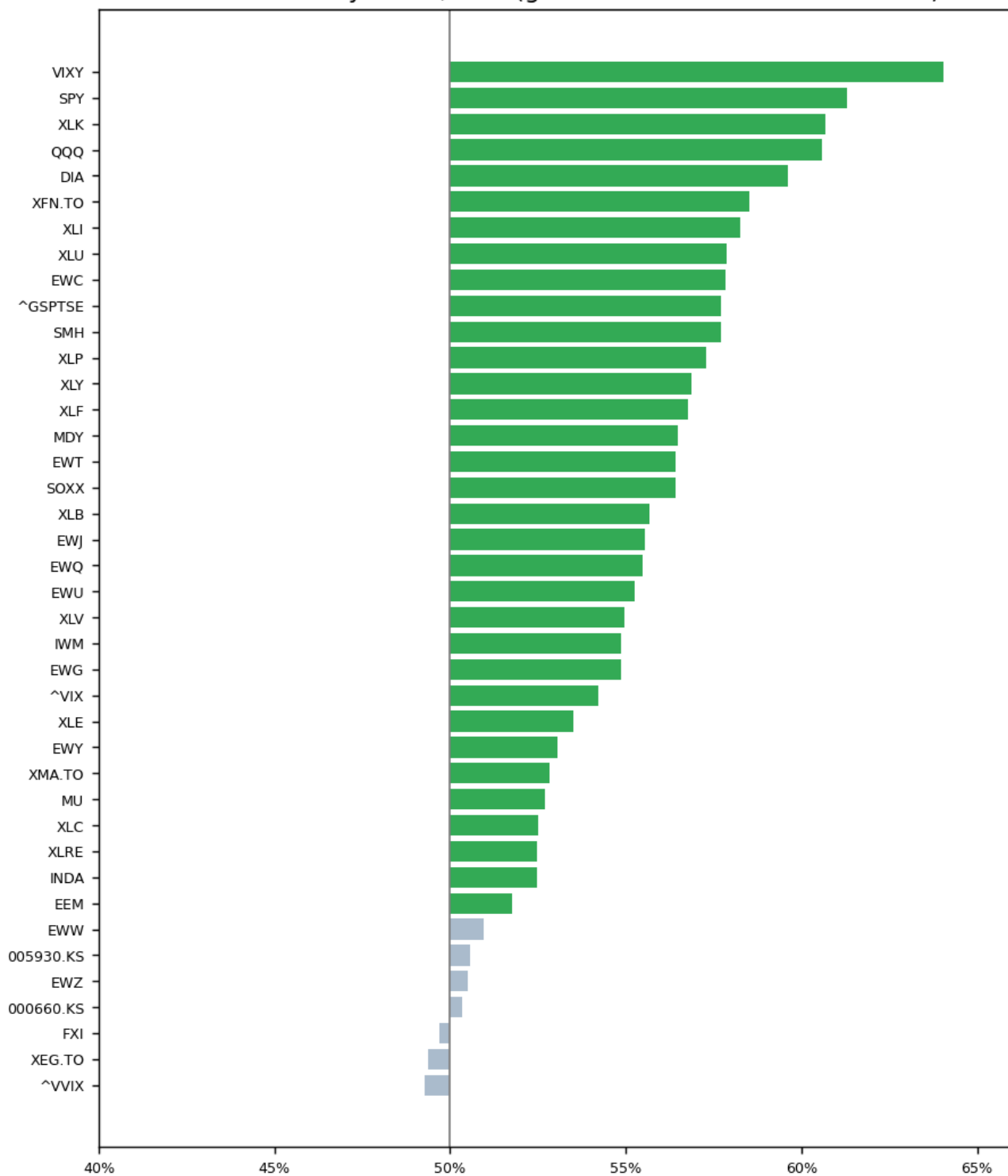
126-day rolling OOS hit rate pooled across tickers,  $h=5$



Reliability, champion series,  $h=5$  (128426 OOS preds)



OOS hit rate by ticker, h=5 (green = survives BH-FDR at 10%)



Champion-series summary by horizon (mean across tickers):

| horizon | tickers | n      | hit    | log_loss | brier  | auc    | fdr_pass |
|---------|---------|--------|--------|----------|--------|--------|----------|
| 1       | 40      | 128504 | 0.5292 | 0.6915   | 0.2491 | 0.4981 | 34       |
| 2       | 40      | 128499 | 0.5356 | 0.6912   | 0.2489 | 0.4946 | 32       |
| 3       | 40      | 128459 | 0.542  | 0.6916   | 0.2488 | 0.4909 | 33       |



| horizon | tickers | n      | hit    | log_loss | brier  | auc    | fdr_pass |
|---------|---------|--------|--------|----------|--------|--------|----------|
| 4       | 40      | 128466 | 0.5475 | 0.69     | 0.2481 | 0.4881 | 34       |
| 5       | 40      | 128426 | 0.5518 | 0.6894   | 0.2477 | 0.489  | 33       |

## 7. Known limitations

(1) The chain-derived SKEW block starts with a single day of history; its value is unmeasurable for months. (2) IBKR implied vol covers ~15 years for large ETFs and less for younger ones; earlier history has no per-ticker IV, only the index complex. (3) ALFRED vintages typically begin 1996-1998; before that, publication-lag shifts are used. (4) The fast-mode backfill uses reduced budgets (annual refits in the nested search, quarterly in the outer walk-forward); the scheduled monthly full ablation re-runs everything at full budget and results supersede. (5) Sequence models (LSTM/TCN) participate only in the full-budget roster. (6) A ~52-55% pooled hit rate on daily horizons is expected; treat any ticker cell that has not survived BH-FDR as noise. (7) ^GSPTSE has no options and the .KS/.TO names have thinner coverage; their SKEW/VOLOPT blocks degrade to the index-level complex.